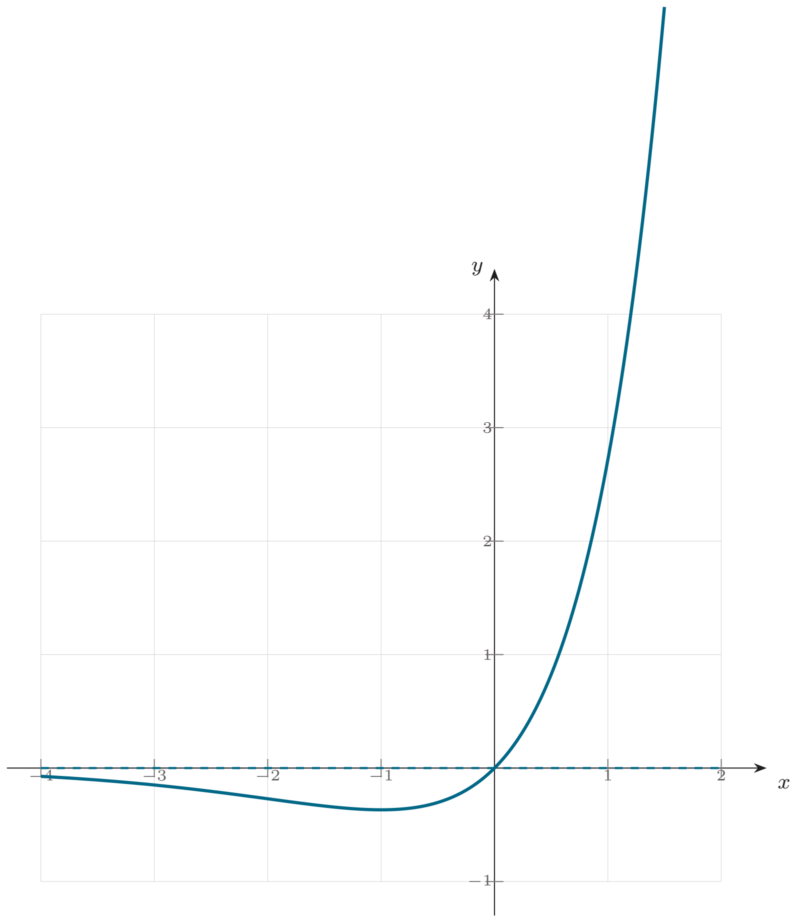
Corrigé — Exercice 14

A. 1) $\lim_{-\infty} g = 0$ (croissance comparée, asymptote $y = 0$) ; $\lim_{+\infty} g = +\infty$. **2)** $g'(x) = \mathrm{e}^x + x\mathrm{e}^x = (x+1)\mathrm{e}^x$: minimum $g(-1) = -\dfrac{1}{\mathrm{e}}$. **3)** $G'(x) = \mathrm{e}^x + (x-1)\mathrm{e}^x = x\mathrm{e}^x = g$. **B. 4)** $I_0 = \int_0^1 \mathrm{e}^x\,\mathrm{d}x = \mathrm{e} - 1$; $I_1 = \int_0^1 x\mathrm{e}^x\,\mathrm{d}x = [G]_0^1 = 0 - (-1) = 1$. **5)** IPP ($u = x^{n+1}$, $\mathrm{d}v = \mathrm{e}^x\,\mathrm{d}x$) : $I_{n+1} = [x^{n+1}\mathrm{e}^x]_0^1 - (n+1)I_n = \mathrm{e} - (n+1)I_n$. **6)** $I_{n+1} - I_n = \int_0^1 x^n(x-1)\mathrm{e}^x\,\mathrm{d}x \leq 0$; intégrande ≥ 0. **7)** $\mathrm{e}^x \leq \mathrm{e}$ sur $[0,1]$ donne $I_n \leq \dfrac{\mathrm{e}}{n+1}$, d'où $\lim I_n = 0$; et $(n+1)I_n = \mathrm{e} - I_{n+1} \to \mathrm{e}$.



$\mathcal{C}_g$: $y = x\mathrm{e}^x$ et l'asymptote $y = 0$ (Exercice 14).